

PETRONAS Net Zero 2050 Pathway - Operations Brief

Document ID	SUS-BRF-014
Version	2.0
Owner	Group Operational Excellence
Classification	Internal

Executive Summary

PETRONAS has committed to achieving Net Zero Carbon Emissions (NZCE) by 2050 across operated assets. This brief summarises the pathway, the operational levers, and the data infrastructure required to track progress.

The pathway is anchored on four levers: (1) operational efficiency, (2) electrification of operations, (3) carbon capture, utilisation and storage (CCUS), and (4) low-carbon fuels and feedstocks.

Scope and Boundaries

Scope 1: Direct emissions from operated assets - the primary focus of operational reporting and the scope tracked in the operations_daily Delta table.

Scope 2: Emissions from purchased electricity, steam, heat or cooling consumed by operated assets.

Scope 3: Indirect emissions across the value chain - addressed through portfolio decisions and customer collaboration.

Interim Targets

By 2030: 25% reduction in absolute Group operational Scope 1+2 GHG emissions versus a 2019 baseline.

By 2040: Methane intensity of less than 0.2% across operated upstream assets.

By 2050: Net zero across all operated Scope 1+2 emissions.

Operational Levers

Energy Efficiency: Targeted 15% improvement in energy intensity by 2030, primarily through compressor optimisation, heat integration and digital advanced process control.

Flaring Reduction: Zero routine flaring by 2030 in line with the World Bank Zero Routine Flaring initiative.

Electrification: Selective electrification of offshore platforms and refinery utilities where renewable power is technically and commercially viable.

CCUS: Build CCUS capacity at the Kasawari and Pengerang complexes to abate hard-to-decarbonise emissions.

Data and Reporting

All operational sites contribute to the Group emissions ledger via the `operations_daily.emissions_tco2e` column. Emission factors are maintained centrally and refreshed annually following external assurance.

The Group's Net Zero performance dashboard is published monthly on Databricks AI/BI Dashboards and shared with the Sustainability Council.

Frequently Asked Questions

Q: What is the Group's Scope 1 reduction target by 2030? A: 25% absolute reduction versus a 2019 baseline, covering operated assets across all business units.

Q: Does Net Zero 2050 cover Scope 3? A: The headline 2050 commitment covers Scope 1 and Scope 2. Scope 3 is addressed through portfolio strategy.

Q: How is methane intensity measured? A: Methane intensity is the volume of methane emissions divided by the volume of marketed natural gas, expressed as a percentage.

This document is synthetic and prepared solely for the PETRONAS AI Enterprise Adoption Workshop. It does not represent actual operational policies of PETRONAS or its subsidiaries.